JULIEN RABAULT

AI Engineer, LLM agents and RAG in production

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I design agent harnesses and ship LLM agents and RAG systems that run in production at customer sites, after four years of applied research at CNRS training and fine-tuning deep learning models on a national supercomputer (PyTorch, multi-GPU). Two peer-reviewed publications.

SKILLS

Agentic systems: Agent harnesses (agent loop, middleware, memory, human-in-the-loop), LangChain / LangGraph / Deep Agents, agentic RAG, MCP, evals (datasets, golden sets, LLM-as-judge), observability, agent security, inference cost tracking, multimodal (OCR, vision, audio), fine-tuning

Languages & data: Python, FastAPI, SQL, Weaviate, PostgreSQL, MongoDB, Hugging Face, Mistral and OpenAI APIs

Infrastructure: Docker, GitLab CI, AWS, Celery, RabbitMQ, S3, Airflow, MLflow, Langfuse, Slurm, Linux

Machine learning: PyTorch, Transformers, multi-GPU training (DDP), diffusion models, computer vision, U-Net, YOLO, GAN, VAE

PROFESSIONAL EXPERIENCE

AI Engineer, Berger-Levrault, Toulouse (AI R&D team)

Jan. 2026 - present

Design and delivery of Athena, Berger-Levrault's agentic platform: agents wired into the document corpora and business APIs of the group's clients (local government, legal, industry, maintenance, public-sector HR). 100k technical documents indexed, 70 pilot users, Langfuse observability, monitoring and cost tracking, client workshops.

- **Athena agent harness** (LangChain / LangGraph), architecture designed and led. Moved from a router (one agent per task: RAG, MCP APIs, reports) to a single agent loop configured per profile and extended through middleware: persistent memory, human approval of sensitive actions, context management (summarisation, token budget), guardrails, skill management, sub-agent orchestration and a unified event contract. Adding a tool no longer means building a new graph. Migration to Deep Agents (LangChain) under way.
- **Agent tools**: agentic RAG, web search, calculator and the matching MCP servers, all plugged into the same loop.
- **Agent evaluation and monitoring**: rebuilt monitoring on Langfuse v4 (actionable traces, dataset management, automated evaluations and scoring), golden sets, LLM-as-judge scoring, comparative benchmarks between architectures and regression tracking.
- **Agent security**: sandboxing of retrieved content against indirect prompt injection, execution bounds on tools (timeouts, per-profile restrictions).
- **MCP Builder**: turns business-unit APIs into MCP servers. An LLM selects the useful endpoints, audits their gaps and generates purpose-built tools (workflows or Python code via FastMCP), human-in-the-loop.
- **Content-extractor**, multimodal extraction (OCR / PDF / DOCX / audio): service built from scratch, 7 extraction pipelines, async batch processing (Celery + Mistral batch API), extensible factory/registry architecture. Deployed, -50% on extraction costs.
- **Document template agent**: tools that extract templates from Word documents, then a second agent and tools that fill those templates from natural language. Shipped to production.
- **LLM pipeline for business data structuring** (public-sector HR): turns interview write-ups into normalised, deduplicated and clustered training needs, then matches them against a catalogue.
- **RAG ingestion chain** (Airflow): added enrichment stages (data augmentation, chunking, embeddings, indexed questions and keywords, time filters). One DAG per client in production.

Machine Learning Engineer, CNRS (PNRIA), Toulouse

Dec. 2021 - Jan. 2026 (4 yrs)

Network of AI engineers supporting research teams (weather, astrophysics, materials, ethology, biology). 10+ projects supported, up to two in parallel (6-12 months), for Météo France, CNES, CEA and INEE. Training and fine-tuning on the Jean Zay supercomputer (multi-GPU DDP, 8 GPUs, Slurm).

- **GENS / MetScore, Météo France:** multi-GPU optimisation and fine-tuning of a diffusion model (DDPM) in PyTorch; designed MetScore (metrics library), still in production. Diffusion POC at -20% compute for equivalent quality. Co-author, AIES 2025.
- **DeepFaune, CNRS / INEE:** fine-tuned YOLOv8 on 1.5M images (24 classes) with class-imbalance handling. 93% accuracy, 3x faster inference. Peer-reviewed publication.

- **BIGSF, CNES:** tech lead on the rebuild of a galactic-filament image analysis library (U-Net): modular architecture, tests, documentation. Public toolbox.
- **AUTOFILL, CEA:** implemented the PairVAE model for generating and completing nanomaterial data, 0.98 mean absolute error. Library parameterised for other materials.
- **MORPHOGAN, Univ. Lorraine:** full rebuild of a StyleGAN2 codebase to study the morphological variability of butterfly wings: automated pipeline, tests, containerisation.
- **Teaching:** created and delivered "Introduction to LLMs" to 25 PhD students and researchers.

Software Engineer, Agileo Automation, Montauban — Aug. 2020 - Sept. 2021 (1 yr)

- **Supervision framework for robotic manufacturing equipment** (semiconductors): C#, object-oriented architecture, HMI, CI/CD. Team of 5, Agile / Scrum.

EDUCATION

MSc Artificial Intelligence & Pattern Recognition, Université Paul Sabatier / IRIT — 2019 - 2021
BSc Computer Science, Université Paul Sabatier Toulouse III — 2016 - 2019

PUBLICATIONS

Enriching Operational High-Resolution Ensemble Forecasts with StyleGAN-2, AIES 2025, peer-reviewed.

The DeepFaune initiative: automatic identification of European fauna, peer-reviewed.

OPEN SOURCE

LangChain: merged contribution to the Mistral integration, surfacing citation metadata.

LLMock (PyPI): LLM mock server for testing retries and fallbacks, 10+ providers.

DDPM-weather: diffusion model for weather image denoising.